Experiment 4: Online Train Reservation System

Aim

To design and develop an online train reservation website using HTML, CSS and JavaScript in which a passenger can log in, search trains by route and date, view seat availability, book tickets for several passengers by class, pay online or offline, receive a PNR number and e-ticket, check PNR status (confirmed / RAC / waiting list) and cancel tickets with refund rules; and the administrator can maintain train details, fares (by zone, class and date), refund rules, payments and refunds.

Algorithm

1. Start and display the login page (Login and Registration). Registration validates name, email, 10-digit phone, password and unique username.
2. Login: verify credentials for the selected role (Passenger / Administrator) and open the matching menu.
3. Passenger - Search: read origin, destination and journey date; list trains whose route (origin, via stations, destination) contains the origin before the destination, with departure/arrival times and via stations.
4. For every class (SL, 3A, 2A) compute availability = seats - passengers already booked for that train, date and class: AVAILABLE n, then RAC (2 berths), then WL. Fare = latest fare rule matching the zone, class and journey date.
5. Passenger details: enter name, age and gender for up to 4 passengers.
6. Payment: Online (validate 16-digit card, MM/YY expiry, 3-digit CVV) or Offline (pay at counter). On success generate a 10-digit PNR and the e-ticket, store payment details for the admin and send the confirmation email.
7. Status allocation: passengers in booking order are given CNF/seat number, then RAC/n, then WL/n.
8. PNR status: enter the PNR number and display the train, journey details and the status of every passenger. The ticket can be printed.
9. Cancellation: refund % = rule for the number of days between today and the journey date; refund = fare x %; status becomes Cancelled and the waiting list is promoted automatically; the refund request goes to the admin.
10. Admin - Trains: add, update or delete train details, route, times, zone and seats per class.
11. Admin - Fares and refund rules: add fares by zone, class and date range; edit refund rules (days before journey and refundable %).
12. Admin - View payment details, bookings, and approve each refund (amount may be adjusted, not above the fare paid); the amount is credited to the passenger.
13. Logout returns to the login page. Stop.

Program (HTML, CSS and JavaScript) - train.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
<title>Online Train Reservation System</title>
<style>
:root{box-sizing:border-box;padding-top:env(safe-area-inset-top,0px);padding-bottom:env(safe-area-inset-bottom,0px);
--bg:#f1f1f8;--card:#fff;--tx:#1e1e33;--mut:#66668a;--ac:#3b3f9e;--acx:#fff;--bd:#d6d6e8;--bad:#c0392b;--ok:#1e8449;--
    warn:#b9770e}
@media (prefers-color-scheme:dark){:root:not([data-theme="light"]){--bg:#131320;--card:#1d1d30;--tx:#e8e8f5;--mut:#9a9ac0;--
    ac:#8c90f0;--acx:#10102a;--bd:#38385a;--bad:#ef7b6e;--ok:#58c183;--warn:#f0b24f}}
html{scroll-padding-top:env(safe-area-inset-top,0px)}
*{box-sizing:border-box}
body{margin:0;background:var(--bg);color:var(--tx);font-family:system-ui,Segoe UI,Arial,sans-serif;line-height:1.45}
header{background:var(--ac);color:var(--acx);padding:12px 16px}
header h1{margin:0 0 8px;font-size:1.15rem}
header nav{display:flex;flex-wrap:wrap;gap:4px}
header button{background:transparent;color:inherit;border:1px solid transparent;padding:5px 10px;border-
    radius:5px;cursor:pointer;font-size:.88rem}
header button.on,header button:hover{border-color:currentColor}
main{max-width:1000px;margin:16px auto;padding:0 12px}
.card{background:var(--card);border:1px solid var(--bd);border-radius:10px;padding:16px;margin-bottom:14px}
button.b{background:var(--ac);color:var(--acx);border:0;border-radius:6px;padding:8px 14px;cursor:pointer;font-size:.9rem}
button.a{background:transparent;color:var(--tx);border:1px solid var(--bd)}
button.s{padding:3px 9px;font-size:.8rem}
button:disabled{opacity:.4;cursor:not-allowed}
input,select{padding:8px;border:1px solid var(--bd);border-radius:6px;background:var(--bg);color:var(--tx);font-size:.9rem;max-
    width:100%}
label{font-size:.85rem;color:var(--mut);display:flex;flex-direction:column;gap:3px}
.row{display:flex;gap:10px;flex-wrap:wrap;align-items:flex-end;margin-bottom:10px}
.tw{overflow-x:auto}
table{border-collapse:collapse;width:100%;font-size:.88rem}
th,td{text-align:left;padding:7px 9px;border-bottom:1px solid var(--bd);white-space:nowrap}
th{color:var(--mut)}
h2{margin:0 0 10px;font-size:1.05rem}
.bad{color:var(--bad);font-weight:700}.ok{color:var(--ok);font-weight:700}.warn{color:var(--warn);font-
    weight:700}.mut{color:var(--mut);font-size:.85rem}
.login{max-width:360px;margin:0 auto}.login input,.login select{width:100%;margin-bottom:8px}
.ticket{border:2px dashed var(--ac);padding:14px;border-radius:8px}
@media print{header,nav,button{display:none!important}}
</style>
</head>
<body>
<header><h1>■ Online Train Reservation System</h1><nav id="nav"></nav></header>
<main id="app"></main>
<script>
const $=id=>document.getElementById(id);
const esc=s=>String(s).replace(/[&<>"]/g,c=>({'&':'&amp;','<':'&lt;','>':'&gt;','"':'&quot;'}[c]));
const today=()=>new Date().toISOString().slice(0,10);
const add=(d,n)=>{const x=new Date(d);x.setDate(x.getDate()+n);return x.toISOString().slice(0,10)};
const days=(a,b)=>Math.round((new Date(b)-new Date(a))/864e5);
const msg=t=>alert(t);
const RAC=2, CLS=['SL','3A','2A'];
const CN={SL:'Sleeper (SL)','3A':'AC 3 Tier (3A)','2A':'AC 2 Tier (2A)'};
const D={
 admin:{u:'admin',p:'admin123'},
 users:[{u:'guest',p:'guest123',name:'Anita Rao',email:'anita@mail.com',phone:'9876543210'}],
 trains:[
  {no:'12675',name:'Kovai
    Express',from:'Chennai',to:'Coimbatore',via:['Katpadi','Salem','Erode'],dep:'06:10',arr:'13:15',zone:'SR',SL:4,'3A':3,'2A':2
    },
  {no:'12243',name:'Chennai-Coimbatore
    Shatabdi',from:'Chennai',to:'Coimbatore',via:['Katpadi','Salem'],dep:'14:45',arr:'21:30',zone:'SR',SL:0,'3A':4,'2A':3},
  {no:'16722',name:'Cbe-Chennai Night
    Mail',from:'Coimbatore',to:'Chennai',via:['Erode','Salem','Katpadi'],dep:'22:00',arr:'05:30',zone:'SR',SL:5,'3A':3,'2A':2}],
 fares:[{zone:'SR',cls:'SL',from:'2000-01-01',to:'2100-12-31',fare:350},{zone:'SR',cls:'3A',from:'2000-01-01',to:'2100-12-
    31',fare:950},{zone:'SR',cls:'2A',from:'2000-01-01',to:'2100-12-31',fare:1400},
        {zone:'SR',cls:'3A',from:'2026-11-01',to:'2026-11-20',fare:1150}],
 rules:[{days:7,pct:90},{days:2,pct:50},{days:0,pct:25}],
 bk:[]
};
let role=null,me=null,view='login',q={},flow=null,tk=null,editT=null,pnrQ='';

const go=v=>{view=v;flow=null;tk=null;render()};
const train=n=>D.trains.find(t=>t.no==n)||{name:'(deleted)',from:'',to:'',via:[],dep:'',zone:''};
const route=t=>[t.from,...t.via,t.to];
const fareOf=(zone,cls,date)=>{const r=[...D.fares].reverse().find(f=>f.zone==zone&&f.cls==cls&&f.from<=date&&date<=f.to);return
     r?r.fare:0};
const act=(tn,d,c)=>D.bk.filter(b=>b.train==tn&&b.date==d&&b.cls==c&&b.st=='Active');
const used=(tn,d,c)=>act(tn,d,c).reduce((s,b)=>s+b.pax.length,0);
function reassign(tn,d,c){
 const cap=train(tn)[c];let i=0;
 act(tn,d,c).forEach(b=>b.pax.forEach(p=>{p.st=i<cap?'CNF/'+(i+1):i<cap+RAC?'RAC/'+(i-cap+1):'WL/'+(i-cap-RAC+1);i++}));
}
function availTxt(tn,d,c){
 const cap=train(tn)[c],u=used(tn,d,c);
 return u<cap?`<span class="ok">AVAILABLE ${cap-u}</span>`:u<cap+RAC?`<span class="warn">RAC ${u-cap+1}</span>`:`<span
    class="bad">WL ${u-cap-RAC+1}</span>`;
}
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const pnrGen=()=>String(Math.floor(2e9+Math.random()*7e8));
function mk(u,tn,d,c,pax,mode){
 const b={pnr:pnrGen(),u,train:tn,date:d,cls:c,pax:pax.map(p=>({...p,st:''})),fare:fareOf(train(tn).zone,c,d),mode,pay:mode=='On
    line'?'Paid':'Pay at counter',st:'Active',bkd:today(),refund:null};
 b.amt=b.fare*b.pax.length;D.bk.push(b);reassign(tn,d,c);return b;
}
mk('ravi','12675',add(today(),5),'SL',[1,2,3,4,5,6].map(i=>({n:'Passenger '+i,a:30,g:'M'})),'Online');

function render(){
 let n=[];
 if(role=='user')n=[['book','Book Ticket'],['my','My Bookings'],['pnr','PNR Status']];
 if(role=='admin')n=[['trains','Trains'],['fares','Fares'],['rules','Refund
    Rules'],['pays','Payments'],['refs','Refunds'],['abk','Bookings']];
 n.push(me?['out','Logout ('+esc(me.u)+')']:['login','Login / Register']);
 $('nav').innerHTML=n.map(x=>`<button class="${x[0]==view?'on':''}"
    onclick="${x[0]=='out'?'logout()':`go('${x[0]}')`}">${x[1]}</button>`).join('');
 const V={login:loginV,book:bookV,my:myV,pnr:pnrV,trains:trainsV,fares:faresV,rules:rulesV,pays:paysV,refs:refsV,abk:abkV};
 $('app').innerHTML=V[view]();
}
function logout(){role=me=null;go('login')}

/* ---- login / register ---- */
function loginV(){
 return `<div class="login"><div class="card"><h2>Login</h2><select id="lr"><option value="user">Passenger</option><option
    value="admin">Administrator</option></select>
 <input id="lu" placeholder="Username"><input id="lp" type="password" placeholder="Password"><button class="b"
    onclick="login()">Login</button>
 <p class="mut">Demo: guest / guest123 | admin / admin123</p></div>
 <div class="card"><h2>Register</h2><input id="rn" placeholder="Full name"><input id="re" placeholder="Email"><input id="rh"
    placeholder="Phone (10 digits)">
 <input id="ru" placeholder="Username"><input id="rp" type="password" placeholder="Password (min 6)"><button class="b"
    onclick="reg()">Register</button></div></div>`;
}
function login(){
 const r=$('lr').value,u=$('lu').value.trim(),p=$('lp').value;
 const f=r=='admin'?(D.admin.u==u&&D.admin.p==p?D.admin:null):D.users.find(x=>x.u==u&&x.p==p);
 if(!f)return msg('Invalid username or password');
 role=r;me=f;go(r=='admin'?'trains':'book');
}
function reg(){
 const
    o={name:$('rn').value.trim(),email:$('re').value.trim(),phone:$('rh').value.trim(),u:$('ru').value.trim(),p:$('rp').value};
 if(!o.name||!o.u)return msg('Please fill all fields');
 if(!/^\S+@\S+\.\S+$/.test(o.email))return msg('Enter a valid email');
 if(!/^\d{10}$/.test(o.phone))return msg('Phone must be 10 digits');
 if(o.p.length<6)return msg('Password must have at least 6 characters');
 if(D.users.some(x=>x.u==o.u))return msg('Username already taken');
 D.users.push(o);msg('Registered successfully. Please login.');
}

/* ---- passenger: search, passengers, payment, ticket ---- */
const stations=()=>[...new Set(D.trains.flatMap(route))];
function bookV(){
 if(tk)return ticketV(tk);
 if(flow)return flowV();
 const s=stations().map(x=>`<option>${x}</option>`).join('');
 let h=`<div class="card"><h2>Search Trains</h2><div class="row"><label>From<select
    id="sf">${s}</select></label><label>To<select id="st">${s}</select></label>
 <label>Date of journey<input type="date" id="sd" min="${today()}" value="${q.d||add(today(),5)}"></label><button class="b"
    onclick="srch()">Search</button></div></div>`;
 if(q.go){
  const l=D.trains.filter(t=>{const r=route(t),i=r.indexOf(q.f),j=r.indexOf(q.t);return i>=0&&j>i});
  h+=`<div class="card tw"><h2>Trains from ${q.f} to ${q.t} on
    ${q.d}</h2><table><tr><th>Train</th><th>Via</th><th>Dep</th><th>Arr</th><th>Class: availability and fare</th></tr>`+
  (l.map(t=>`<tr><td><b>${t.no}</b> ${esc(t.name)}</td><td>${esc(t.via.join(', '))}</td><td>${t.dep}</td><td>${t.arr}</td><td>`+
   CLS.filter(c=>t[c]>0).map(c=>`<div style="margin:3px 0">${c}: ${availTxt(t.no,q.d,c)} | Rs. ${fareOf(t.zone,c,q.d)} <button
    class="b s" onclick="pick('${t.no}','${c}')">Book</button></div>`).join('')+'</td></tr>').join('')||'<tr><td colspan="5"
    class="mut">No trains on this route.</td></tr>')+'</table></div>';
 }
 return h;
}
function srch(){
 const f=$('sf').value,t=$('st').value,d=$('sd').value;
 if(f==t)return msg('Origin and destination cannot be the same');
 if(!d||d<today())return msg('Select a valid journey date');
 q={go:1,f,t,d};render();$('sf').value=f;$('st').value=t;
}
function pick(no,c){flow={no,c,d:q.d,f:q.f,t:q.t,stage:'pax',n:1,pax:[]};render()}
function flowV(){
 const t=train(flow.no),fare=fareOf(t.zone,flow.c,flow.d);
 const head=`<div class="card"><h2>${t.no} ${esc(t.name)} - ${CN[flow.c]}</h2><div>${flow.f} → ${flow.t} | ${flow.d} | Dep
    ${t.dep} | Fare Rs. ${fare} per passenger | ${availTxt(t.no,flow.d,flow.c)}</div></div>`;
 if(flow.stage=='pax'){
  let r='';for(let i=0;i<flow.n;i++){const p=flow.pax[i]||{};r+=`<div class="row"><b>${i+1}.</b><label>Name<input id="pn${i}"
    value="${esc(p.n||'')}"></label>
   <label>Age<input id="pa${i}" type="number" min="1" max="110" style="width:80px"
    value="${p.a||''}"></label><label>Gender<select id="pg${i}"><option ${p.g=='M'?'selected':''}>M</option><option
    ${p.g=='F'?'selected':''}>F</option></select></label></div>`}
  return head+`<div class="card"><h2>Passenger Details (max 4)</h2>${r}<button class="b a" onclick="addP()">+ Add
    passenger</button> <button class="b" onclick="toPay()">Continue to Payment</button> <button class="b a"
    onclick="go('book')">Back</button></div>`;
 }
 return head+`<div class="card login" style="max-width:440px"><h2>Payment</h2><p>Total fare: <b>Rs. ${fare*flow.pax.length}</b>
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    for ${flow.pax.length} passenger(s)</p>
 <select id="pm"
    onchange="$('cd').style.display=this.value=='Online'?'block':'none'"><option>Online</option><option>Offline</option></select
    >
 <div id="cd"><input id="cn" placeholder="Card number (16 digits)"><input id="cx" placeholder="MM/YY"><input id="cv"
    type="password" placeholder="CVV" maxlength="3"></div>
 <p class="mut">Offline: pay at the reservation counter within 24 hours.</p><button class="b" onclick="payNow()">Confirm and
    Pay</button> <button class="b a" onclick="flow.stage='pax';render()">Back</button></div>`;
}
function readP(){for(let i=0;i<flow.n;i++){flow.pax[i]={n:($('pn'+i)||{value:''}).value.trim(),a:+($('pa'+i)||{value:0}).value,g
    :($('pg'+i)||{value:'M'}).value}}}
function addP(){readP();if(flow.n>=4)return msg('Maximum 4 passengers per ticket');flow.n++;render()}
function toPay(){
 readP();
 if(flow.pax.some(p=>!p.n||!(p.a>0)))return msg('Enter name and age for every passenger');
 flow.stage='pay';render();
}
function payNow(){
 const m=$('pm').value;
 if(m=='Online'){
  if(!/^\d{16}$/.test($('cn').value.replace(/\s/g,'')))return msg('Card number must be 16 digits');
  if(!/^(0[1-9]|1[0-2])\/\d{2}$/.test($('cx').value))return msg('Expiry must be MM/YY');
  if(!/^\d{3}$/.test($('cv').value))return msg('CVV must be 3 digits');
 }
 const b=mk(me.u,flow.no,flow.d,flow.c,flow.pax,m);
 msg('Ticket generated. PNR: '+b.pnr+'. Payment details sent to admin; confirmation emailed to '+me.email+'.');
 flow=null;tk=b;render();
}
function ticketV(b){
 const t=train(b.train);
 return `<div class="card"><div class="ticket"><h2>E-Ticket - PNR ${b.pnr}</h2><p><b>${b.train} ${esc(t.name)}</b> |
    ${CN[b.cls]}<br>${t.from} → ${t.to} | Journey: ${b.date} | Dep ${t.dep}<br>Booked on ${b.bkd} | Payment: ${b.mode}
    (${b.pay})</p>
 <table><tr><th>#</th><th>Name</th><th>Age/Gender</th><th>Status</th></tr>${b.pax.map((p,i)=>`<tr><td>${i+1}</td><td>${esc(p.n)}
    </td><td>${p.a}/${p.g}</td><td>${stTxt(p.st)}</td></tr>`).join('')}</table>
 <p><b>Total fare: Rs. ${b.amt}</b></p></div><p><button class="b" onclick="window.print()">Print Ticket</button> <button
    class="b a" onclick="go('my')">My Bookings</button></p></div>`;
}
const stTxt=s=>`<span
    class="${s.startsWith('CNF')?'ok':s.startsWith('RAC')?'warn':s.startsWith('WL')?'bad':''}">${s||'Cancelled'}</span>`;

/* ---- passenger: history, cancel, PNR ---- */
function myV(){
 if(tk)return ticketV(tk);
 return `<div class="card tw"><h2>My Bookings</h2><table><tr><th>PNR</th><th>Train</th><th>Date</th><th>Class</th><th>Passengers
    </th><th>Amount</th><th>Status</th><th>Refund</th><th></th></tr>`+
 (D.bk.filter(b=>b.u==me.u).map(b=>`<tr><td>${b.pnr}</td><td>${b.train}
    ${esc(train(b.train).name)}</td><td>${b.date}</td><td>${b.cls}</td><td>${b.pax.length}</td><td>Rs. ${b.amt}</td>
 <td class="${b.st=='Active'?'ok':'bad'}">${b.st}</td><td>${b.refund?`Rs. ${b.refund.amt} (${b.refund.status})`:'-'}</td>
 <td><button class="b a s" onclick="tk=D.bk.find(x=>x.pnr=='${b.pnr}');render()">View/Print</button> ${b.st=='Active'?`<button
    class="b a s" onclick="cancelB('${b.pnr}')">Cancel</button>`:''}</td></tr>`).join('')||'<tr><td colspan="9" class="mut">No
    bookings yet.</td></tr>')+'</table></div>';
}
function pct(date){const r=[...D.rules].sort((a,b)=>b.days-a.days).find(x=>days(today(),date)>=x.days);return r?r.pct:0}
function cancelB(pnr){
 const b=D.bk.find(x=>x.pnr==pnr);
 if(b.date<today())return msg('Journey date has passed');
 const p=pct(b.date),amt=Math.round(b.amt*p/100);
 if(!confirm('Cancel PNR '+pnr+'? Refundable: '+p+'% = Rs. '+amt))return;
 b.st='Cancelled';b.refund={pct:p,amt,status:'Pending'};b.pax.forEach(x=>x.st='');
 reassign(b.train,b.date,b.cls);msg('Cancellation request accepted. Refund of Rs. '+amt+' will be released after admin
    approval.');render();
}
function pnrV(){
 const b=pnrQ&&D.bk.find(x=>x.pnr==pnrQ);
 return `<div class="card"><h2>Check PNR Status</h2><div class="row"><input id="pq" placeholder="Enter 10-digit PNR"
    value="${esc(pnrQ)}"><button class="b" onclick="pnrQ=$('pq').value.trim();render()">Check</button></div></div>`+
 (pnrQ?(b?ticketV(b).replace('<p><button class="b" onclick="window.print()">Print Ticket</button> <button class="b a"
    onclick="go(\'my\')">My Bookings</button></p>',''):'<div class="card bad">No booking found for this PNR.</div>'):'');
}

/* ---- admin ---- */
function trainsV(){
 const e=train(editT);const has=!!D.trains.find(t=>t.no==editT);
 const v=has?e:{no:'',name:'',from:'',to:'',via:[],dep:'',arr:'',zone:'SR',SL:0,'3A':0,'2A':0};
 return `<div class="card"><h2>${has?'Update':'Add'} Train</h2><div class="row">
 <label>Train no<input id="tn" value="${v.no}" style="width:90px" ${has?'readonly':''}></label><label>Name<input id="tm"
    value="${esc(v.name)}"></label>
 <label>Origin<input id="tf" value="${esc(v.from)}" style="width:110px"></label><label>Destination<input id="tt"
    value="${esc(v.to)}" style="width:110px"></label>
 <label>Via (comma separated)<input id="tv" value="${esc(v.via.join(', '))}"></label><label>Dep<input id="td" type="time"
    value="${v.dep}"></label><label>Arr<input id="ta" type="time" value="${v.arr}"></label>
 <label>Zone<input id="tz" value="${v.zone}" style="width:60px"></label>
 <label>SL seats<input id="c0" type="number" min="0" value="${v.SL}" style="width:70px"></label><label>3A seats<input id="c1"
    type="number" min="0" value="${v['3A']}" style="width:70px"></label><label>2A seats<input id="c2" type="number" min="0"
    value="${v['2A']}" style="width:70px"></label>
 <button class="b" onclick="saveT()">Save</button>${has?'<button class="b a"
    onclick="editT=null;render()">Cancel</button>':''}</div></div>
 <div class="card tw"><h2>Trains</h2><table><tr><th>No</th><th>Name</th><th>Route</th><th>Dep</th><th>Zone</th><th>SL</th><th>3A
    </th><th>2A</th><th></th></tr>`+
 D.trains.map(t=>`<tr><td>${t.no}</td><td>${esc(t.name)}</td><td>${route(t).join(' →
    ')}</td><td>${t.dep}</td><td>${t.zone}</td><td>${t.SL}</td><td>${t['3A']}</td><td>${t['2A']}</td>
 <td><button class="b a s" onclick="editT='${t.no}';render()">Edit</button> <button class="b a s"
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    onclick="delT('${t.no}')">Delete</button></td></tr>`).join('')+'</table></div>';
}
function saveT(){
 const o={no:$('tn').value.trim(),name:$('tm').value.trim(),from:$('tf').value.trim(),to:$('tt').value.trim(),via:$('tv').value.
    split(',').map(x=>x.trim()).filter(Boolean),
  dep:$('td').value,arr:$('ta').value,zone:$('tz').value.trim(),SL:+$('c0').value,'3A':+$('c1').value,'2A':+$('c2').value};
 if(!/^\d{5}$/.test(o.no))return msg('Train number must be 5 digits');
 if(!o.name||!o.from||!o.to||!o.dep||!o.arr||!o.zone)return msg('Fill all train details');
 const i=D.trains.findIndex(t=>t.no==o.no);
 if(i>=0)D.trains[i]=o;else D.trains.push(o);editT=null;render();
}
function delT(no){if(D.bk.some(b=>b.train==no&&b.st=='Active'&&b.date>=today()))return msg('Train has active
    bookings');D.trains=D.trains.filter(t=>t.no!=no);render()}
function faresV(){
 return `<div class="card"><h2>Add Fare (zone, class, date range)</h2><div class="row"><label>Zone<input id="fz" value="SR"
    style="width:70px"></label>
 <label>Class<select id="fc">${CLS.map(c=>`<option>${c}</option>`).join('')}</select></label><label>From date<input type="date"
    id="ff"></label><label>To date<input type="date" id="fe"></label>
 <label>Fare (Rs.)<input id="fp" type="number" min="1" style="width:90px"></label><button class="b"
    onclick="addFare()">Add</button></div><p class="mut">If date ranges overlap, the most recently added fare applies.</p></div>
 <div class="card tw"><h2>Fare
    Table</h2><table><tr><th>Zone</th><th>Class</th><th>From</th><th>To</th><th>Fare</th><th></th></tr>`+
 D.fares.map((f,i)=>`<tr><td>${f.zone}</td><td>${f.cls}</td><td>${f.from}</td><td>${f.to}</td><td>Rs. ${f.fare}</td><td><button
    class="b a s" onclick="D.fares.splice(${i},1);render()">Delete</button></td></tr>`).join('')+'</table></div>';
}
function addFare(){
 const f={zone:$('fz').value.trim(),cls:$('fc').value,from:$('ff').value,to:$('fe').value,fare:+$('fp').value};
 if(!f.zone||!f.from||!f.to||f.to<f.from||f.fare<1)return msg('Enter zone, valid date range and fare');
 D.fares.push(f);render();
}
function rulesV(){
 return `<div class="card"><h2>Refund Rules</h2><p class="mut">Refund % depends on days between cancellation date and journey
    date.</p><table><tr><th>Days before journey (at least)</th><th>Refundable %</th><th></th></tr>`+
 D.rules.map((r,i)=>`<tr><td><input type="number" min="0" id="rd${i}" value="${r.days}" style="width:80px"></td><td><input
    type="number" min="0" max="100" id="rp${i}" value="${r.pct}" style="width:80px"></td><td><button class="b a s"
    onclick="D.rules.splice(${i},1);render()">Delete</button></td></tr>`).join('')+
 `</table><p><button class="b" onclick="saveRules()">Save Rules</button> <button class="b a"
    onclick="D.rules.push({days:0,pct:0});render()">+ Add rule</button></p></div>`;
}
function saveRules(){D.rules.forEach((r,i)=>{r.days=+$('rd'+i).value;r.pct=Math.min(100,+$('rp'+i).value)});msg('Refund rules
    saved');render()}
function paysV(){
 return `<div class="card tw"><h2>Payment Details (received from
    system)</h2><table><tr><th>PNR</th><th>User</th><th>Train</th><th>Booked on</th><th>Mode</th><th>Amount</th><th>Payment
    status</th></tr>`+
 D.bk.map(b=>`<tr><td>${b.pnr}</td><td>${esc(b.u)}</td><td>${b.train}</td><td>${b.bkd}</td><td>${b.mode}</td><td>Rs.
    ${b.amt}</td><td>${b.pay}</td></tr>`).join('')+'</table></div>';
}
function refsV(){
 const l=D.bk.filter(b=>b.refund);
 return `<div class="card tw"><h2>Cancellations and Refunds</h2><table><tr><th>PNR</th><th>User</th><th>Fare paid</th><th>Refund
     %</th><th>Refund amount (Rs.)</th><th>Status</th><th></th></tr>`+
 (l.map((b,i)=>`<tr><td>${b.pnr}</td><td>${esc(b.u)}</td><td>Rs. ${b.amt}</td><td>${b.refund.pct}%</td><td><input type="number"
    id="ra${b.pnr}" value="${b.refund.amt}" style="width:90px" ${b.refund.status=='Paid'?'disabled':''}></td>
 <td class="${b.refund.status=='Paid'?'ok':'warn'}">${b.refund.status}</td><td>${b.refund.status=='Pending'?`<button class="b s"
     onclick="payRef('${b.pnr}')">Approve and Pay</button>`:''}</td></tr>`).join('')||'<tr><td colspan="7" class="mut">No
    cancellations.</td></tr>')+'</table></div>';
}
function payRef(pnr){
 const b=D.bk.find(x=>x.pnr==pnr),a=+$('ra'+pnr).value;
 if(!(a>=0)||a>b.amt)return msg('Refund cannot exceed the fare paid');
 b.refund.amt=a;b.refund.status='Paid';msg('Refund of Rs. '+a+' credited to passenger '+b.u);render();
}
function abkV(){
 return `<div class="card tw"><h2>All
    Bookings</h2><table><tr><th>PNR</th><th>User</th><th>Train</th><th>Date</th><th>Class</th><th>Passengers
    (status)</th><th>Amount</th><th>State</th></tr>`+
 D.bk.map(b=>`<tr><td>${b.pnr}</td><td>${esc(b.u)}</td><td>${b.train}</td><td>${b.date}</td><td>${b.cls}</td><td>${b.pax.map(p=>
    stTxt(p.st)).join(', ')}</td><td>Rs. ${b.amt}</td><td>${b.st}</td></tr>`).join('')+'</table></div>';
}
render();
</script>
</body>
</html>
```

Output

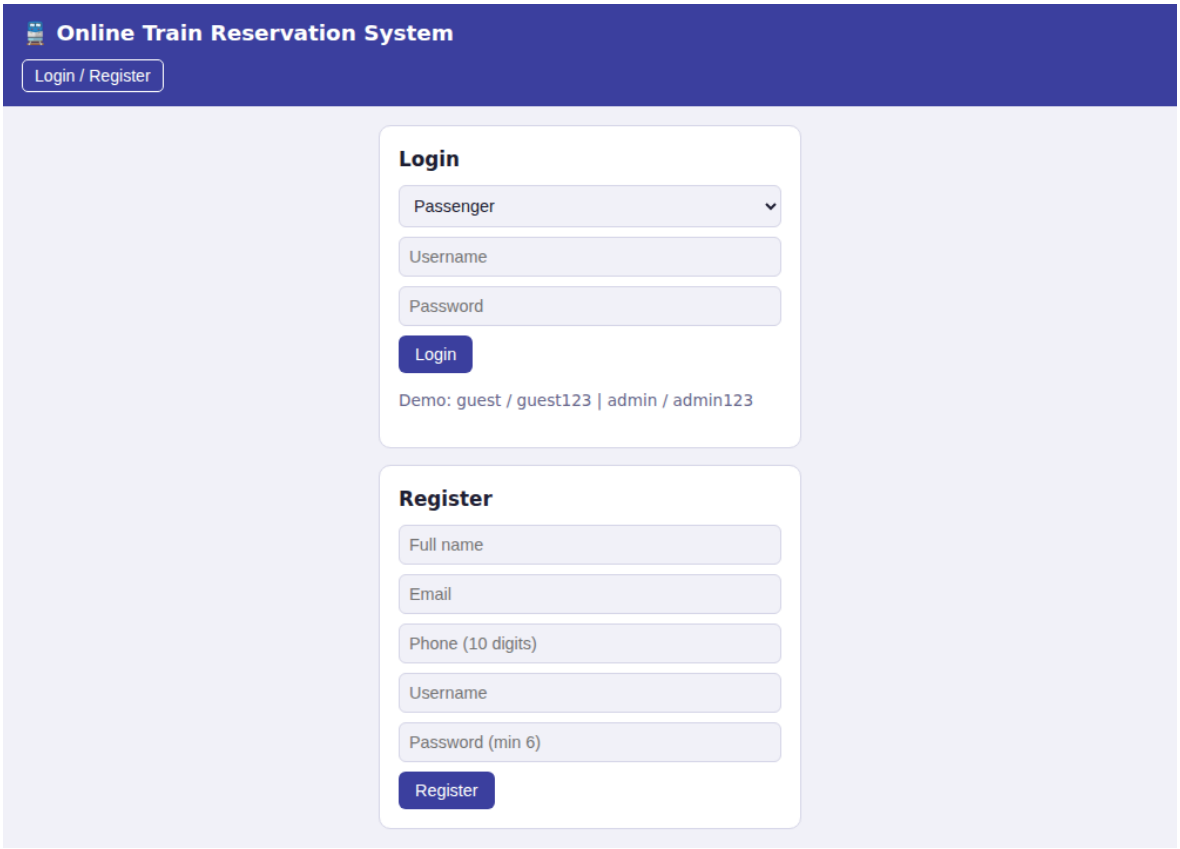


Fig 1: Login and registration page

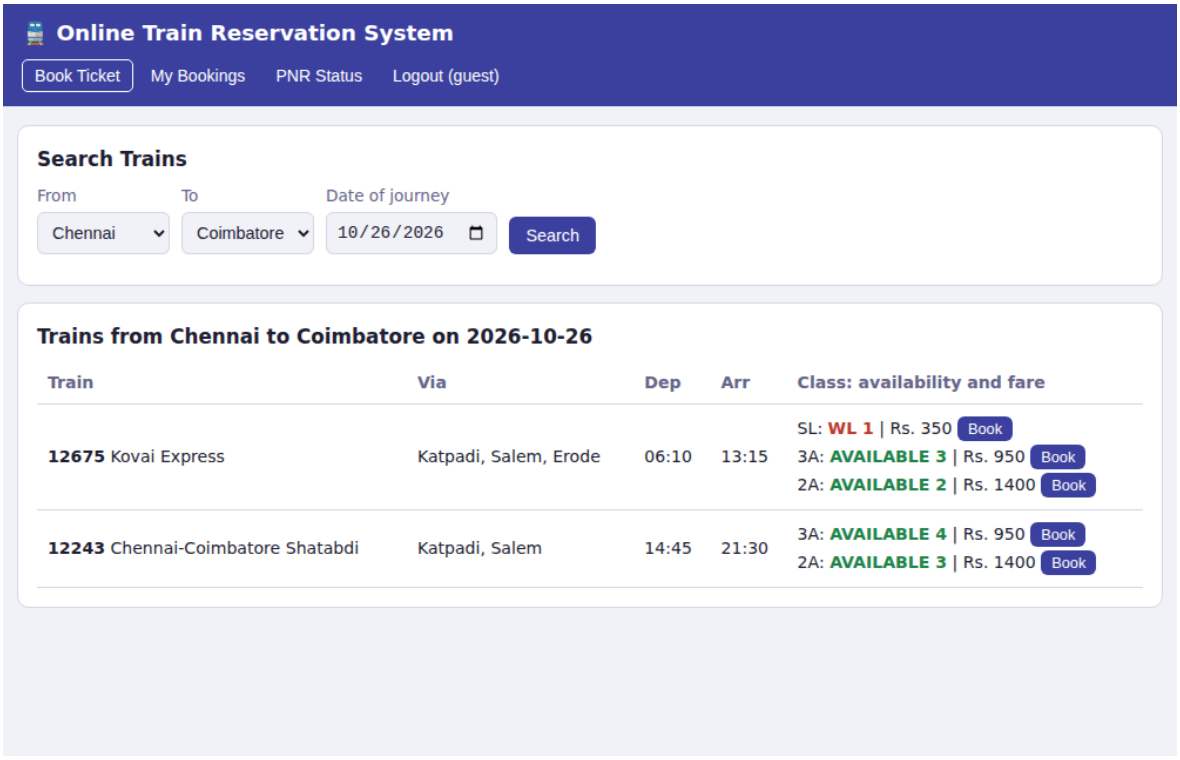


Fig 2: Train list with availability, via stations and fare

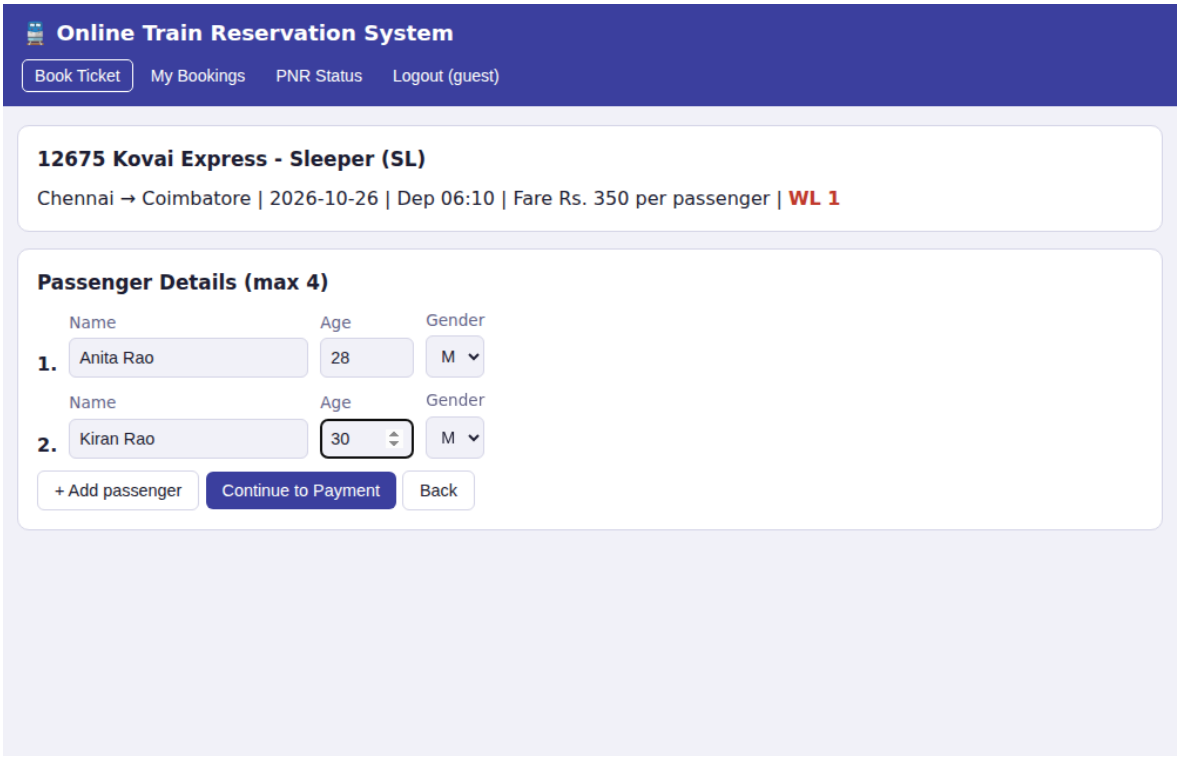


Fig 3: Passenger details

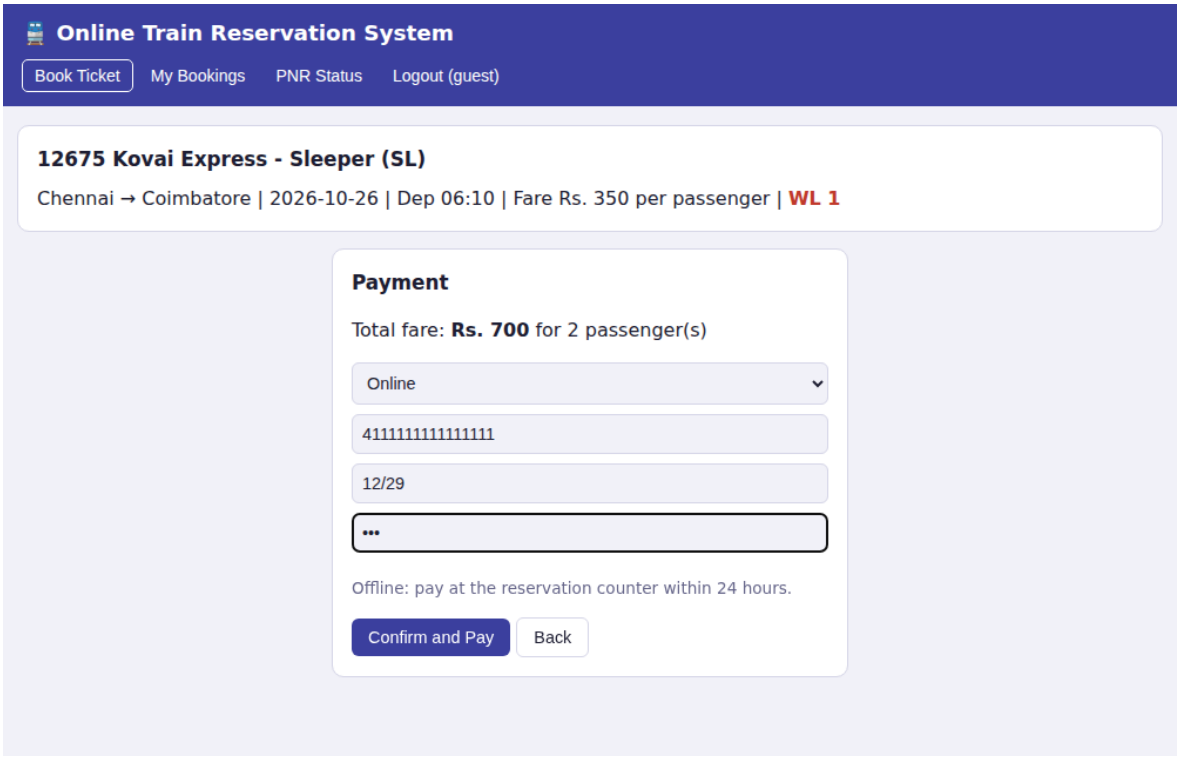


Fig 4: Payment (online / offline)

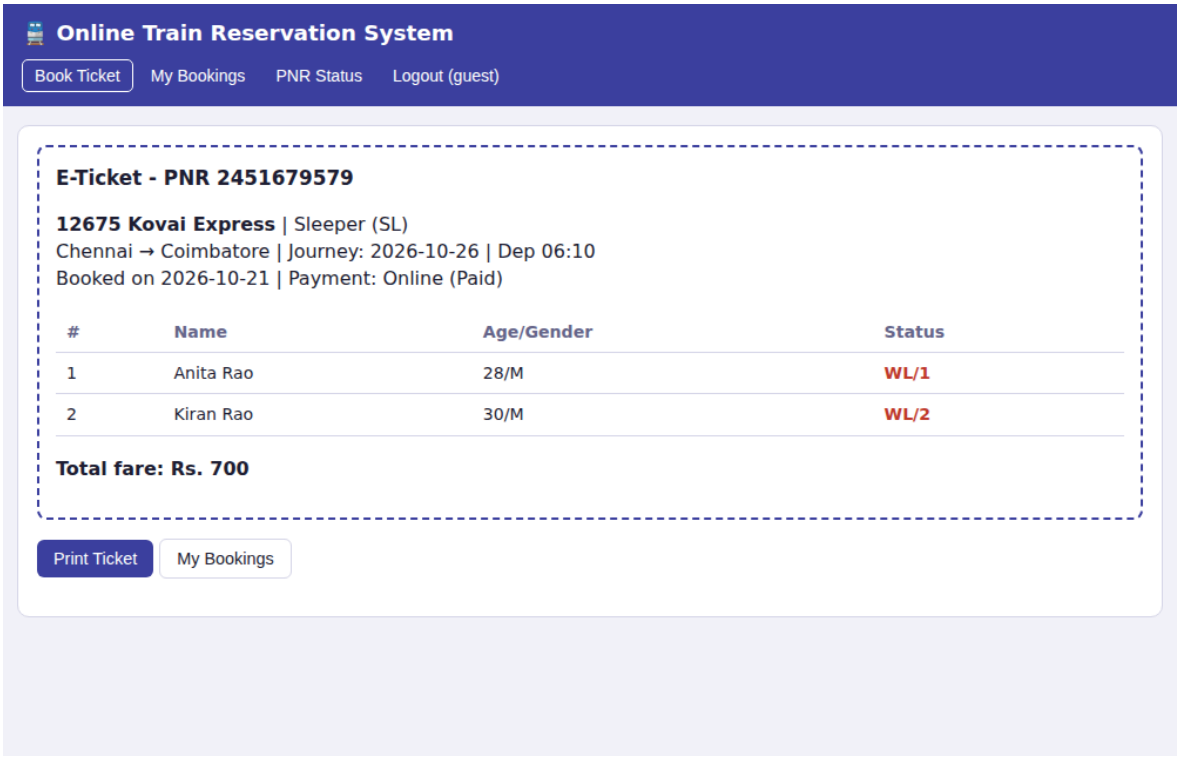


Fig 5: E-ticket with PNR and waiting-list status

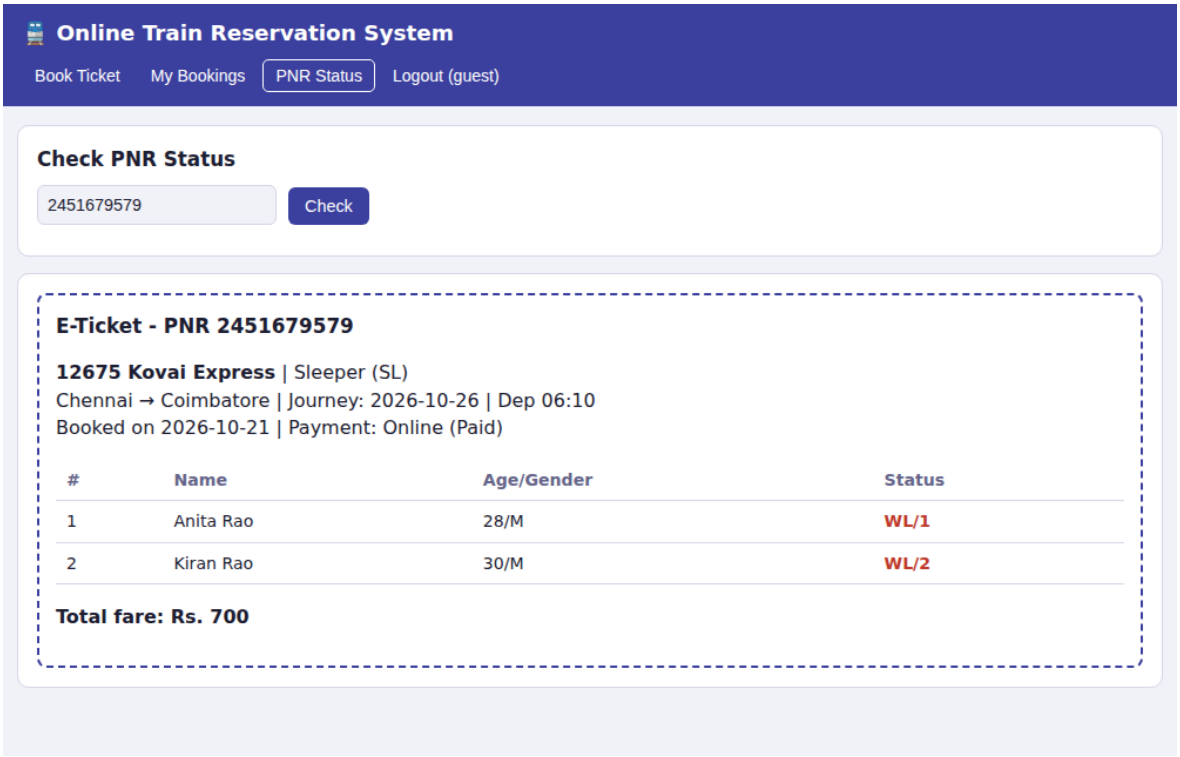


Fig 6: Check PNR status

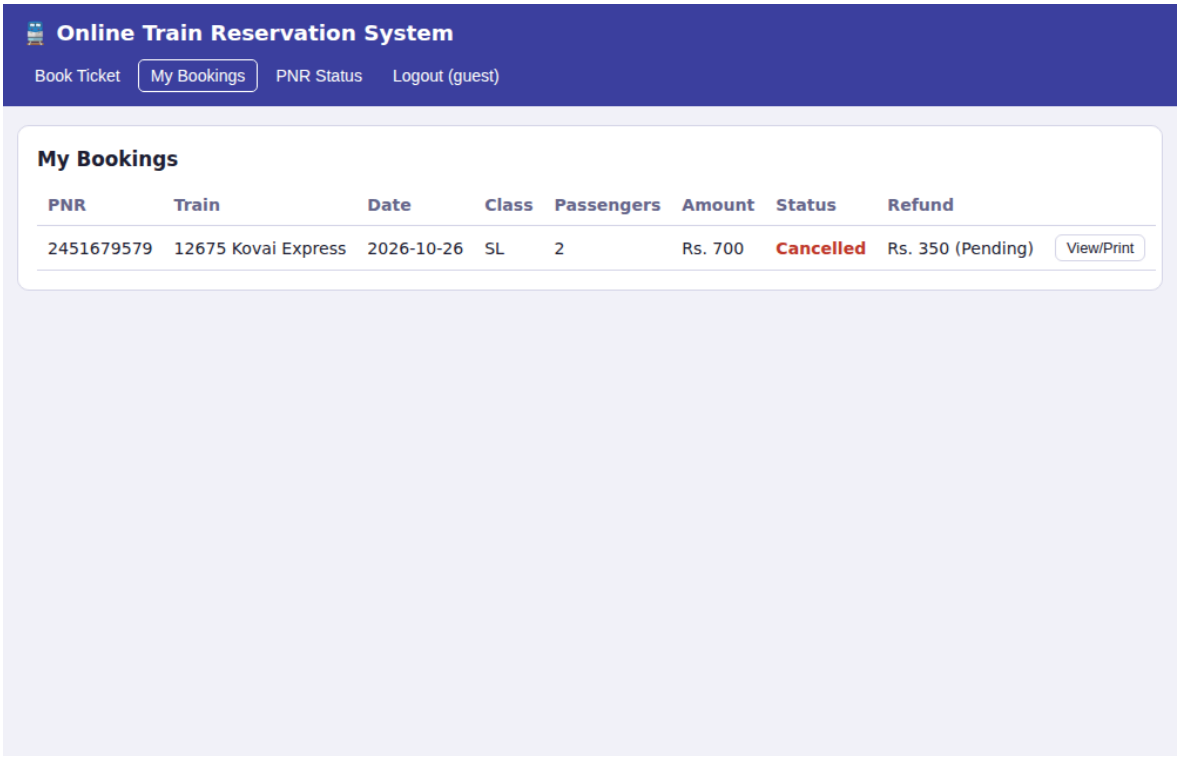


Fig 7: Cancellation - 50% refundable, status Cancelled

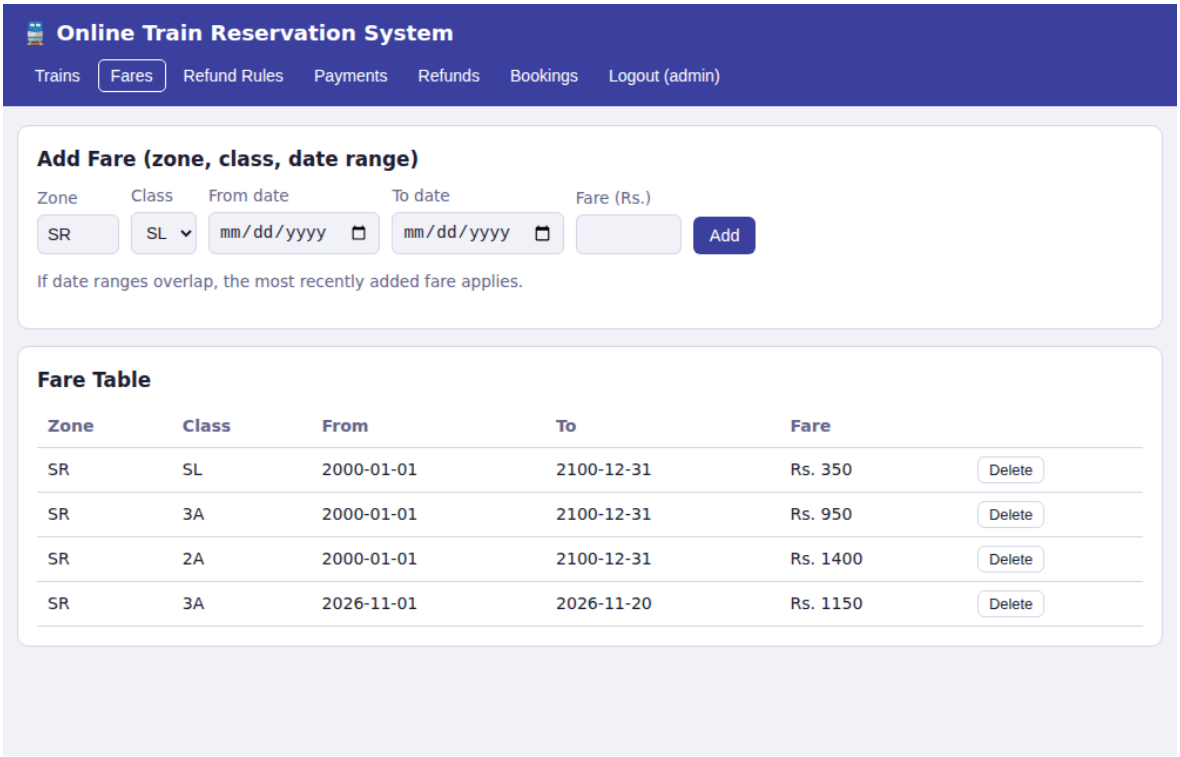


Fig 8: Admin - Fare table (zone, class, date range)

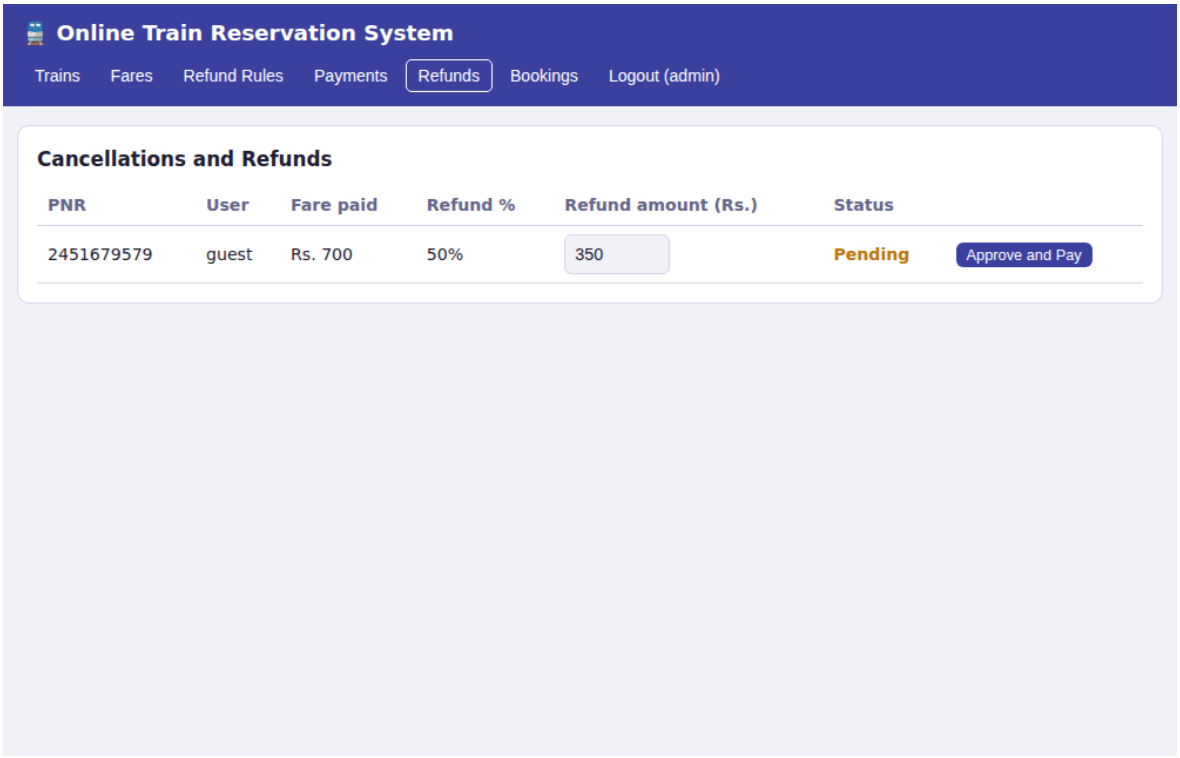


Fig 9: Admin - Refund request pending

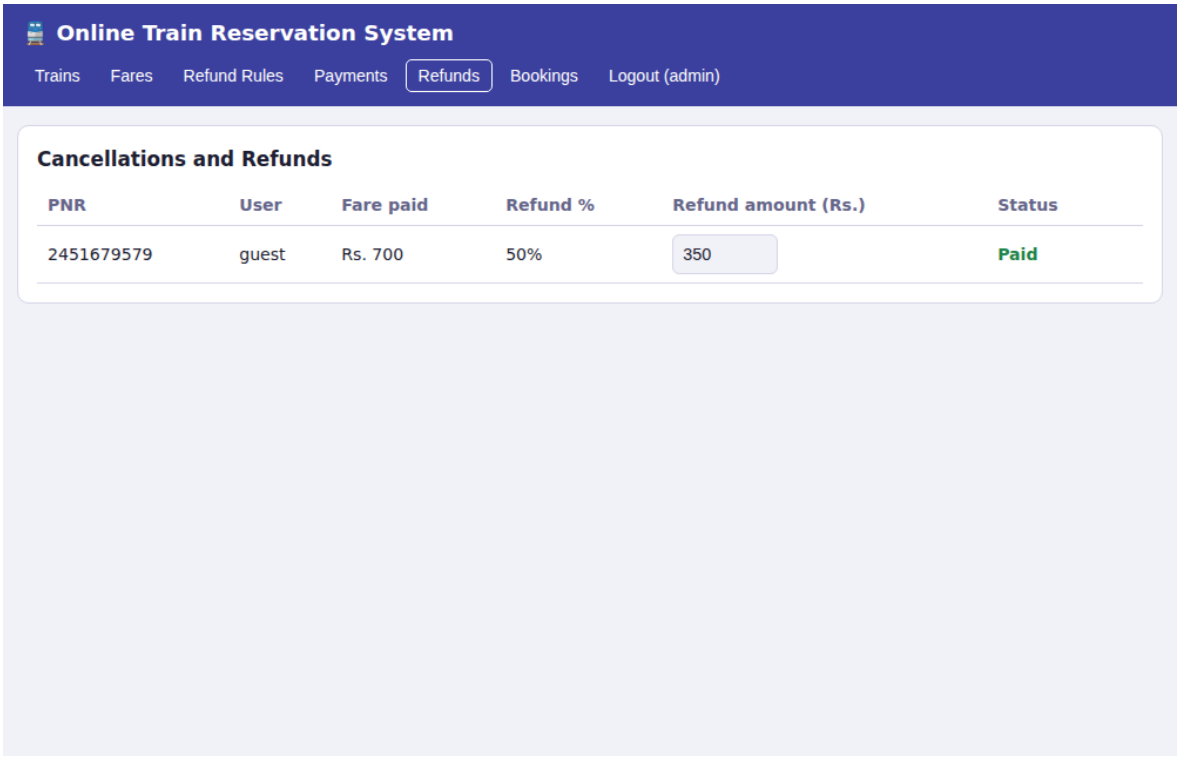


Fig 10: Admin - Refund approved and paid

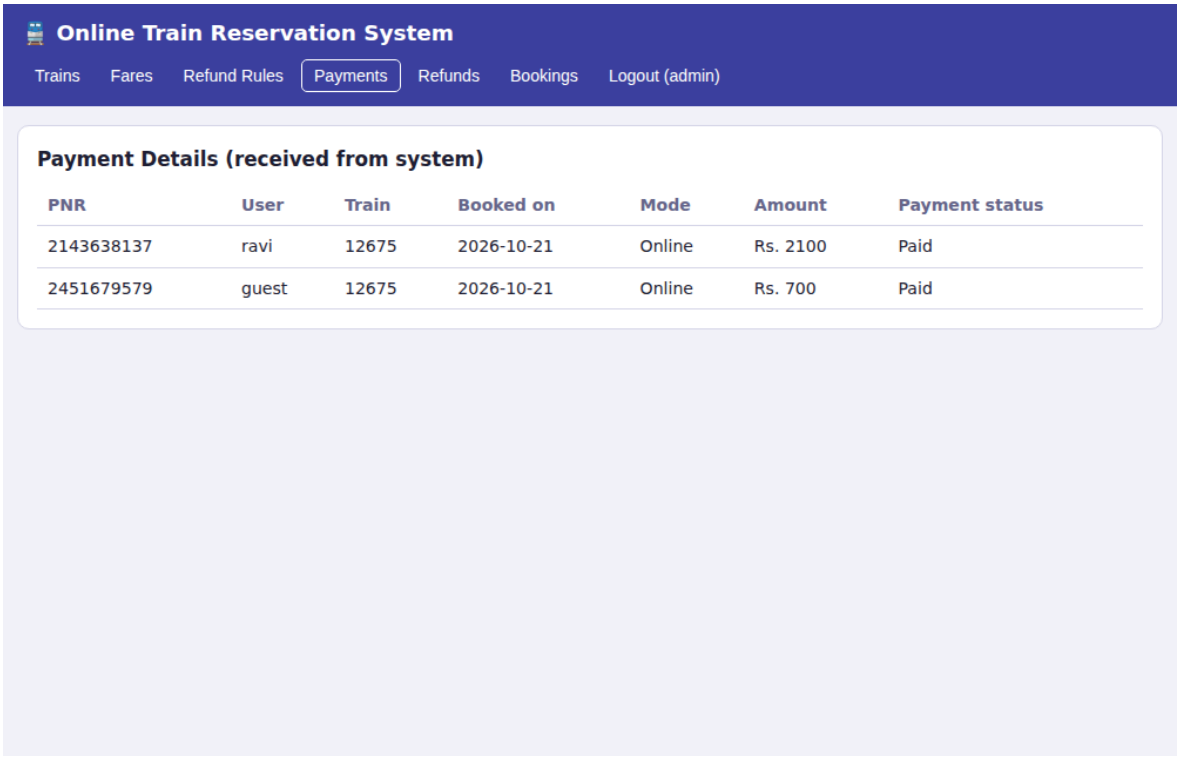


Fig 11: Admin - Payment details

Result

The online train reservation website was developed using HTML, CSS and JavaScript. Passenger operations (login, train search, availability, booking, payment, PNR generation and status, ticket printing, cancellation) and administrator operations (train and fare maintenance, refund rules, payment details, refund approval) were executed successfully.